# 第二十五讲

# 虚功原理与单位载荷法

作业：13-16,17(b),18(a),21,24,26;

## 第十三章　能量法

13.1 外力功与应变能

13.2 功的互等定理

13.3 卡氏定理

13.4 虚功原理与单位载荷法

# 上讲回顾

## 卡氏定理

$$\Delta_k = \frac{\partial V_\varepsilon}{\partial F_k}$$

- 适用于线弹性体；
- 附加载荷；
- 区分载荷；
- 对广义载荷求偏导——求相对位移。

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13.4 虚功原理与单位载荷法

# 第十三章 能量法

# 13.4 虚功原理与单位载荷法

1、变形体虚功原理

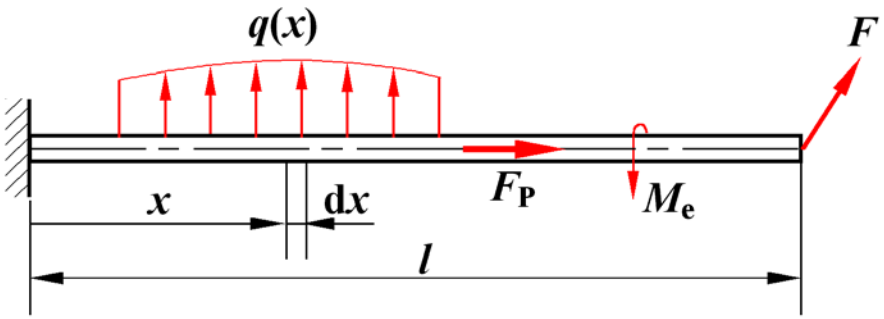
2、虚功原理的例证

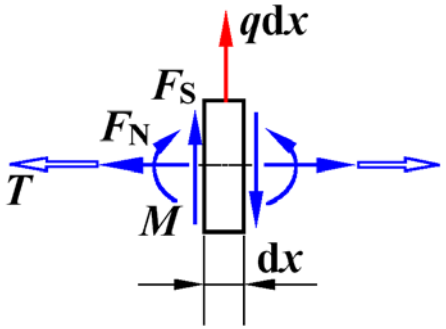
3、单位载荷法

4、单位载荷法应用

# 1、变形体虚功原理

## 1）与外力平衡的可能内力



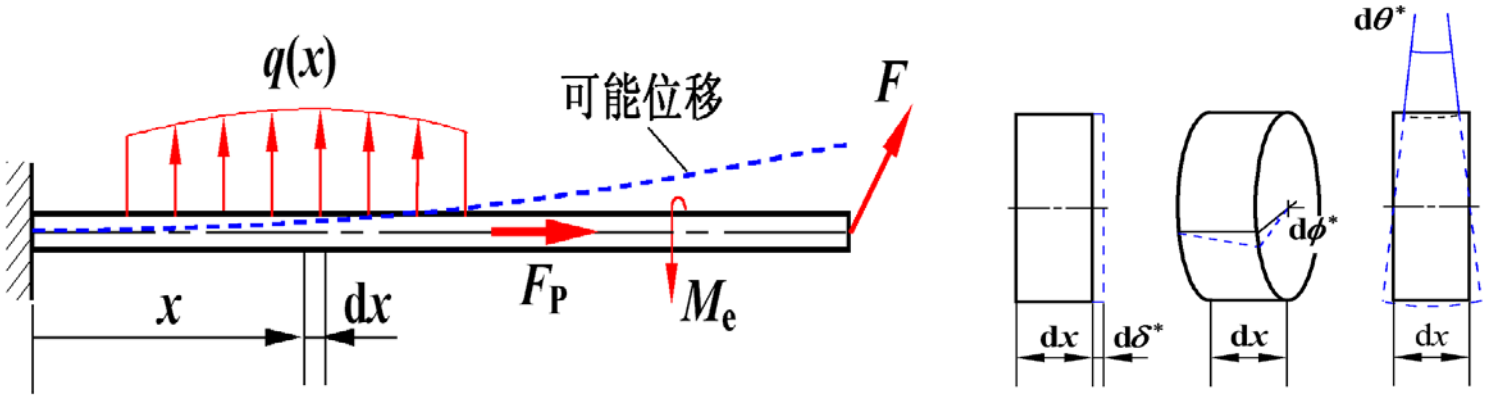

1）与外力保持平衡并满足静力边界条件的内力，称为静力可能内力或可能内力

2）杆的可能内力用$F_N$, $T$, $F_S$与$M$表示

3）可能内力与外力 ⟶ 结构的静力许可场

# 1、变形体虚功原理

## 2）可能位移与可能变形




1）满足变形连续条件与位移边界条件的任意结构位移，称为几何可能位移，相应之变形称为可能变形

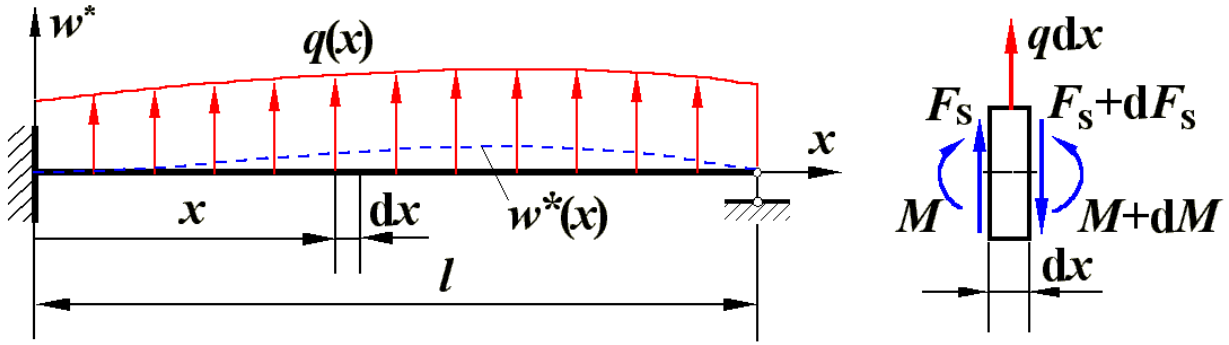
2）杆件的位移：　$\Delta l, \varphi(x), w(x), \theta(x)$

3）杆微段的变形用d$\delta$，d$\phi$与d$\theta$ 表示

运动
许可场

# 1、变形体虚功原理

例如




可能内力满足： $\dfrac{\mathrm{d}F_S}{\mathrm{d}x}=q,\quad \dfrac{\mathrm{d}M}{\mathrm{d}x}=F_S$（平衡条件）

$$M(l)=0$$ （静力边界条件）

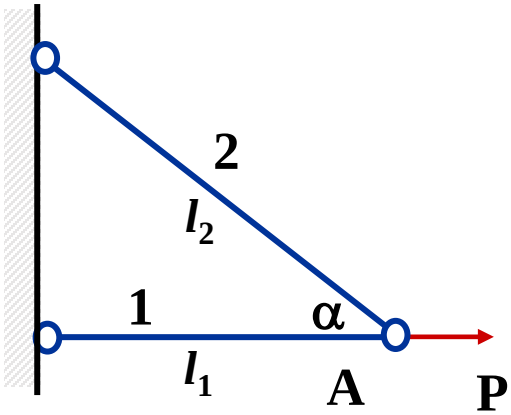
可能位移与变形满足：

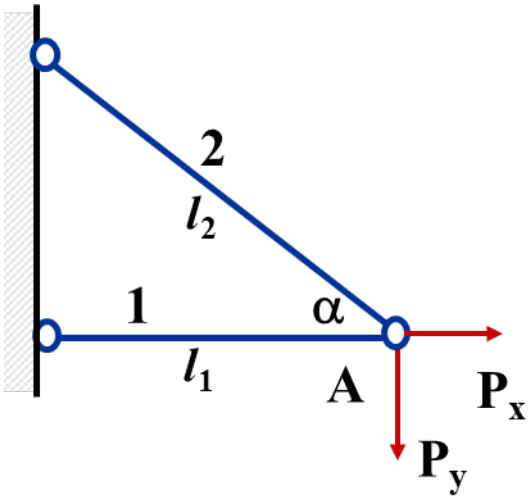
$$\theta^*=\frac{\mathrm{d}w^*}{\mathrm{d}x}$$ （变形连续条件）

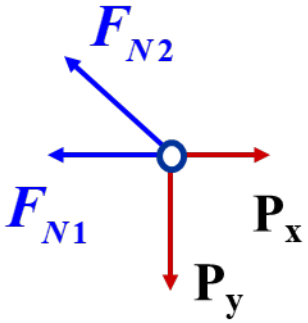
$$w^*(0)=w^*(l)=0,\quad \theta^*(0)=0$$ （位移边界条件）

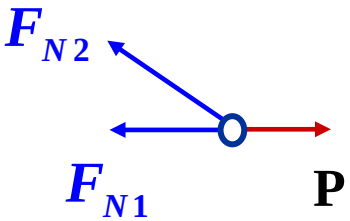
# 1、变形体虚功原理

可能内力



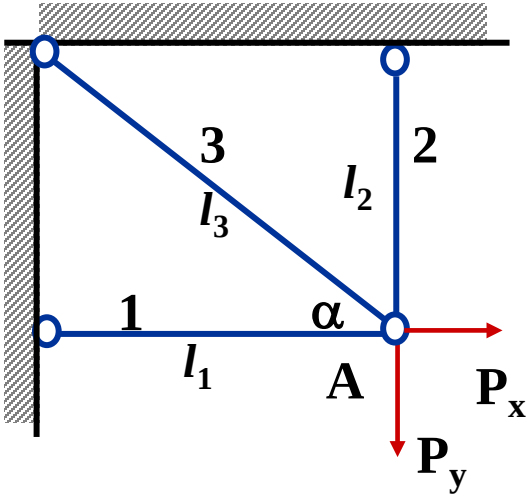

$$F_{N1} = P_x - P_y \frac{\cos\alpha}{\sin\alpha},$$

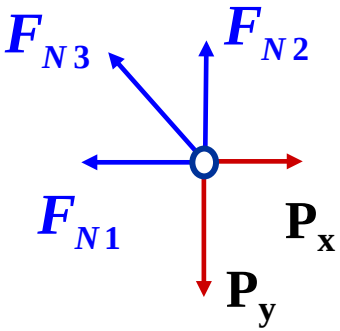
$$F_{N1} = P, \quad F_{N2} = 0$$

$$F_{N2} = \frac{P_y}{\sin\alpha}$$

# 1、变形体虚功原理

可能内力




$$\begin{cases} F_{N1} + F_{N3}\cos\alpha = P_x \\ F_{N2} + F_{N3}\sin\alpha = P_y \end{cases}$$

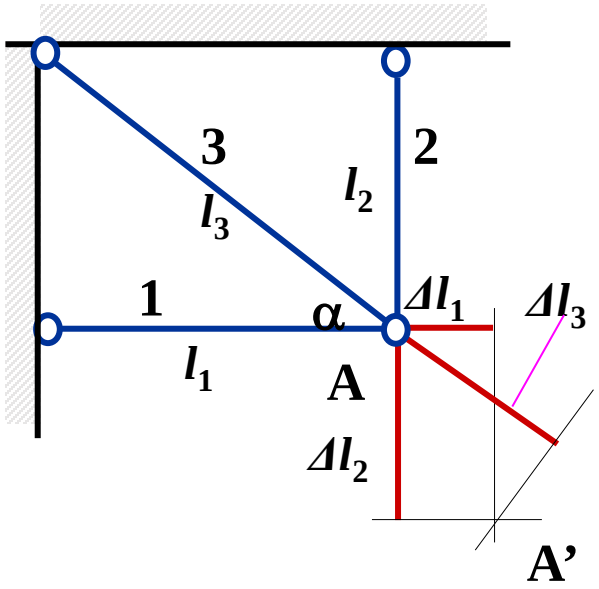
平衡方程有无数内力解答；

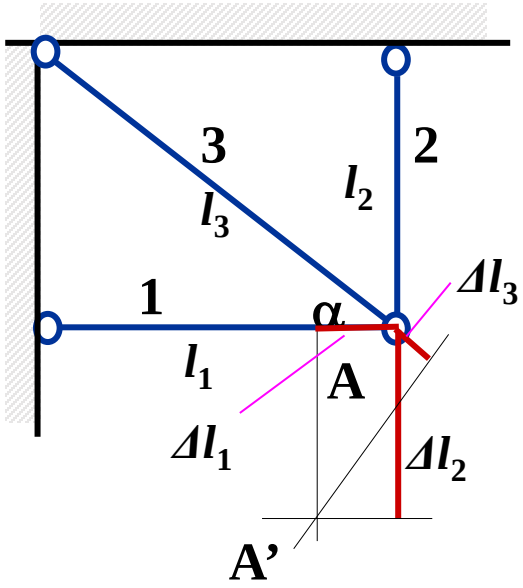
任一组内力解答均为可能内力

静不定结构有无数多可能内力

# 1、变形体虚功原理

可能变形




$$\Delta l_2 \sin\alpha + \Delta l_1 \cos\alpha = \Delta l_3$$

$$\Delta l_2 \sin\alpha - \Delta l_1 \cos\alpha = \Delta l_3$$

满足变形协调条件的任一组变形均为可能变形

# 1、变形体虚功原理

## 3）虚位移与虚变形

- 假想的、微小的可能位移，称为结构的虚位移，
- 相应之变形称为虚变形；
- 杆微段的虚变形用$\mathrm{d}\delta^*$、$\mathrm{d}\phi^*$与$\mathrm{d}\theta^*$表示

## 4）虚功（以梁为例）

如温度变形

$q(x)$ $\quad$ $M(x)$ $\quad$ 结构的外力与内力

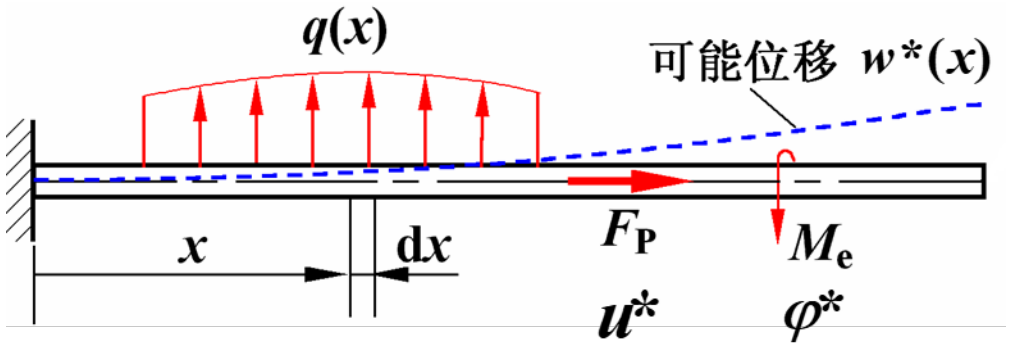
$w^*(x)$ $\quad$ $\mathrm{d}\theta^*$ $\quad$ 任取结构的虚位移与虚变形

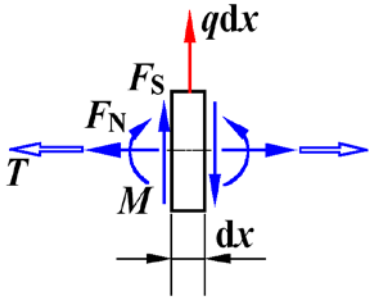
$W_e$ $\quad$ $W_i$

如何表示？
二者的关系？

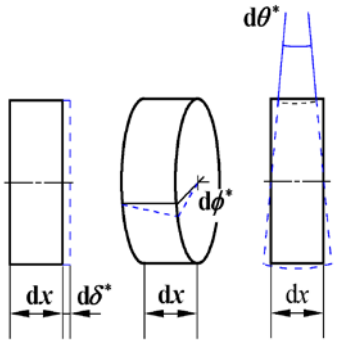
# 1、变形体虚功原理

## 5）内力虚功与外力虚功




❶ 作用在所有微段上的可能内力在虚变形上所作之总虚功—内力虚功




$$W_i = \int_l (F_N \mathrm{d}\delta^* + T\mathrm{d}\phi^* + M\mathrm{d}\theta^*)$$

$$W_i = \int_l (F_N \mathrm{d}\delta^* + T\mathrm{d}\phi^* + M_y \mathrm{d}\theta_y^* + M_z \mathrm{d}\theta_z^*)$$

❷ 外力在可能位移上所作之总虚功—外力虚功

$$W_e = \int_{l_q} q(x) w^*(x) \mathrm{d}x + M_e \varphi^* + F_p u^*$$

# 1、变形体虚功原理

## 6）变形体虚功原理

Principle of Virtual Work

可以证明：

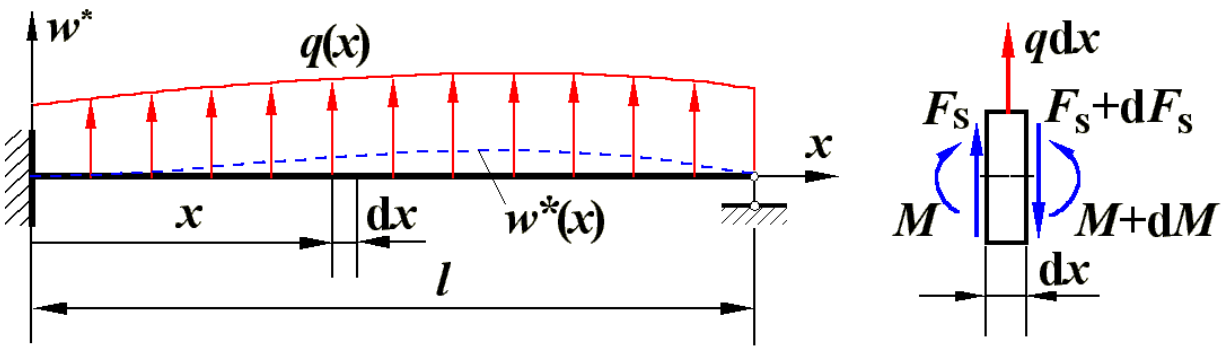
外力在虚位移上所作外力虚功 $W_e$，等于可能内力在虚变形上所作内力虚功 $W_i$，即

$$W_e = W_i$$

称为变形体虚功原理（虚位移原理）

# 2、变形体虚功原理的例证

## 1）一端固定一端铰支梁




可能内力满足：$\dfrac{\mathrm{d}F_S}{\mathrm{d}x}=q,\quad \dfrac{\mathrm{d}M}{\mathrm{d}x}=F_S$ （平衡条件）

$M(l)=0$ （静力边界条件）

虚位移满足：$\theta^*=\dfrac{\mathrm{d}w^*}{\mathrm{d}x}$ （变形连续条件）

$w^*(0)=w^*(l)=0,\quad \theta^*(0)=0$ （位移边界条件）

# 2、变形体虚功原理的例证

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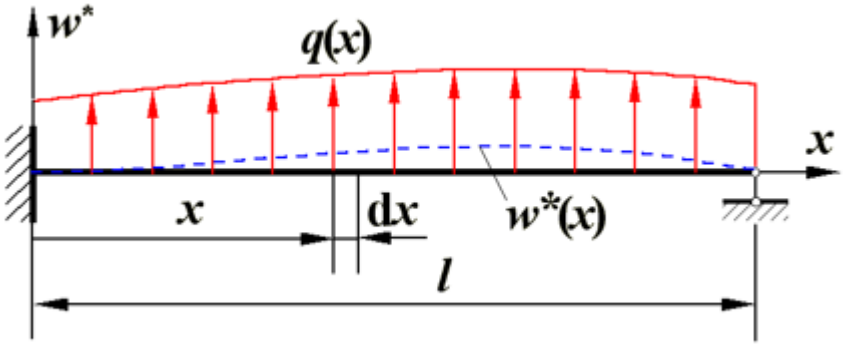



$$\frac{\mathrm{d}F_S}{\mathrm{d}x}=q,\qquad \frac{\mathrm{d}M}{\mathrm{d}x}=F_S\qquad M(l)=0$$

$$\theta^*=\frac{\mathrm{d}w^*}{\mathrm{d}x}\qquad w^*(0)=w^*(l)=0\qquad \theta^*(0)=0$$

外力虚功

$$W_e=\int_l w^*q\mathrm{d}x=\int_l w^*\frac{\mathrm{d}F_S}{\mathrm{d}x}\mathrm{d}x=[F_Sw^*]_0^l-\int_l F_S\frac{\mathrm{d}w^*}{\mathrm{d}x}\mathrm{d}x$$

$$=0-\int_l\frac{\mathrm{d}M}{\mathrm{d}x}\theta^*\mathrm{d}x=-[M\theta^*]_0^l+\int_l M\mathrm{d}\theta^*=\int_l M\mathrm{d}\theta^*$$

内力虚功 $W_i=\int_l M\mathrm{d}\theta^*$

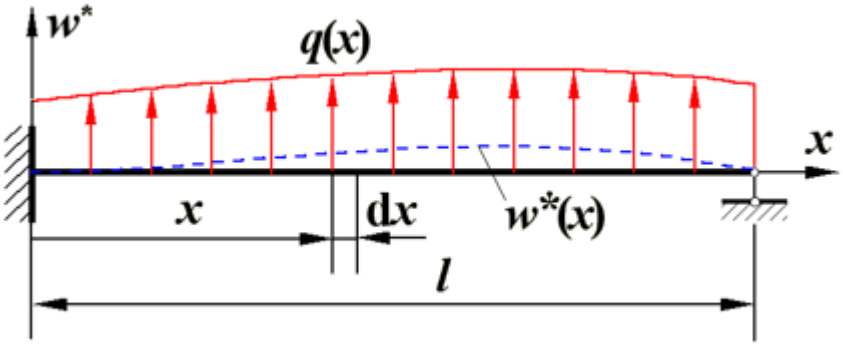
$$\boxed{\therefore\quad W_e=W_i}$$

# 2、变形体虚功原理的例证

$$\frac{\mathrm{d}F_S}{\mathrm{d}x}=q,\qquad \frac{\mathrm{d}M}{\mathrm{d}x}=F_S\qquad M(l)=0$$

$$\theta^*=\frac{\mathrm{d}w^*}{\mathrm{d}x}\qquad w^*(0)=w^*(l)=0$$

$$\theta^*(0)=0$$




## 2）简支梁

位移边界条件与静力边界条件

$$\boxed{w^*(0)=w^*(l)=0\qquad M(l)=0\qquad M(0)=0}$$

$$W_e=\int_l w^*q\mathrm{d}x=\int_l w^*\frac{\mathrm{d}F_S}{\mathrm{d}x}\mathrm{d}x=\left[F_Sw^*\right]_0^l-\int_l F_S\frac{\mathrm{d}w^*}{\mathrm{d}x}\mathrm{d}x$$

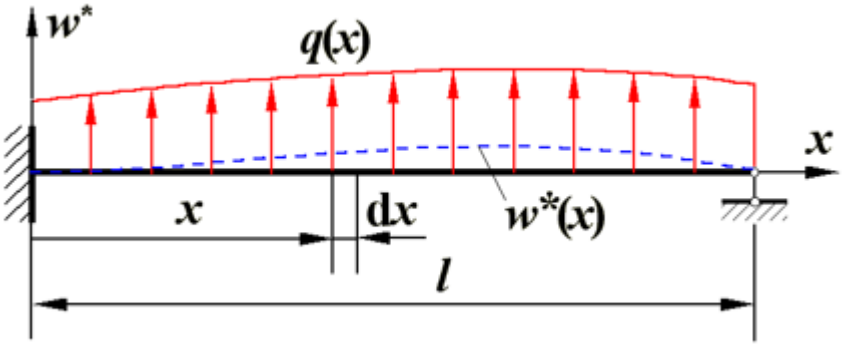
$$=0-\int_l\frac{\mathrm{d}M}{\mathrm{d}x}\theta^*\mathrm{d}x=-\left[M\theta^*\right]_0^l+\int_l M\mathrm{d}\theta^*=\int_l M\mathrm{d}\theta^*=W_i$$

# 2、变形体虚功原理的例证

$$\frac{\mathrm{d}F_S}{\mathrm{d}x}=q,\qquad \frac{\mathrm{d}M}{\mathrm{d}x}=F_S\qquad M(l)=0$$

$$\theta^*=\frac{\mathrm{d}w^*}{\mathrm{d}x}\qquad w^*(0)=w^*(l)=0$$

$$\theta^*(0)=0$$




## 3）悬臂梁

位移边界条件与静力边界条件

$$\boxed{w^*(0)=0\quad \theta^*(0)=0\quad M(l)=0\quad F_S(l)=0}$$

$$W_e=\int_l w^*q\mathrm{d}x=\int_l w^*\frac{\mathrm{d}F_S}{\mathrm{d}x}\mathrm{d}x=\left[F_Sw^*\right]_0^l-\int_l F_S\frac{\mathrm{d}w^*}{\mathrm{d}x}\mathrm{d}x$$

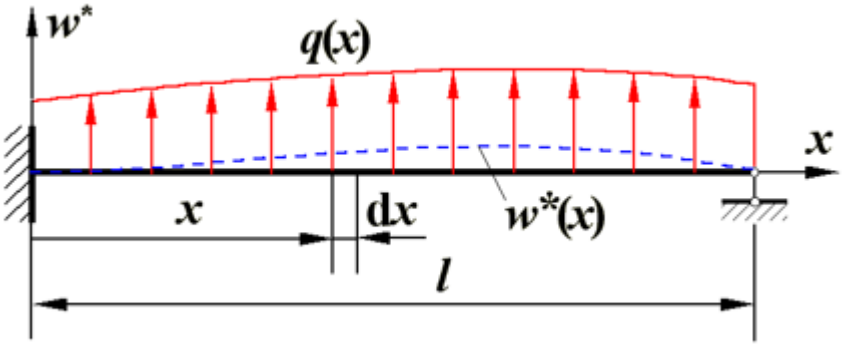
$$=0-\int_l\frac{\mathrm{d}M}{\mathrm{d}x}\theta^*\mathrm{d}x=-\left[M\theta^*\right]_0^l+\int_l M\mathrm{d}\theta^*=\int_l M\mathrm{d}\theta^*=W_i$$

# 2、变形体虚功原理的例证

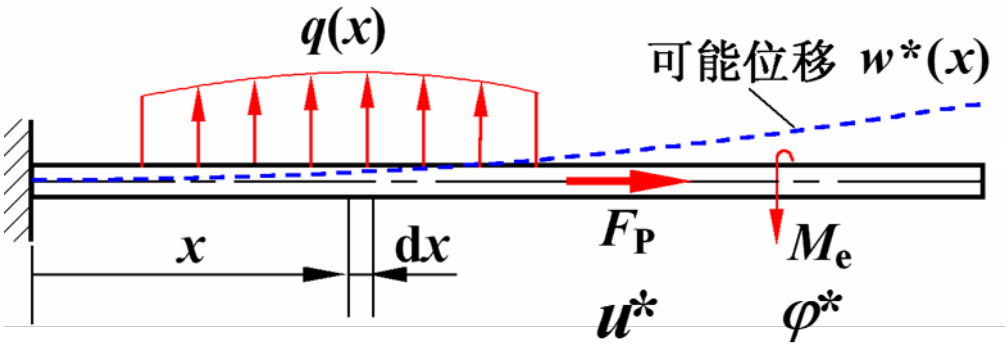
$$\frac{\mathrm{d}F_S}{\mathrm{d}x}=q,\qquad \frac{\mathrm{d}M}{\mathrm{d}x}=F_S\qquad M(l)=0$$

$$\theta^*=\frac{\mathrm{d}w^*}{\mathrm{d}x}\qquad w^*(0)=w^*(l)=0$$

$$\theta^*(0)=0$$



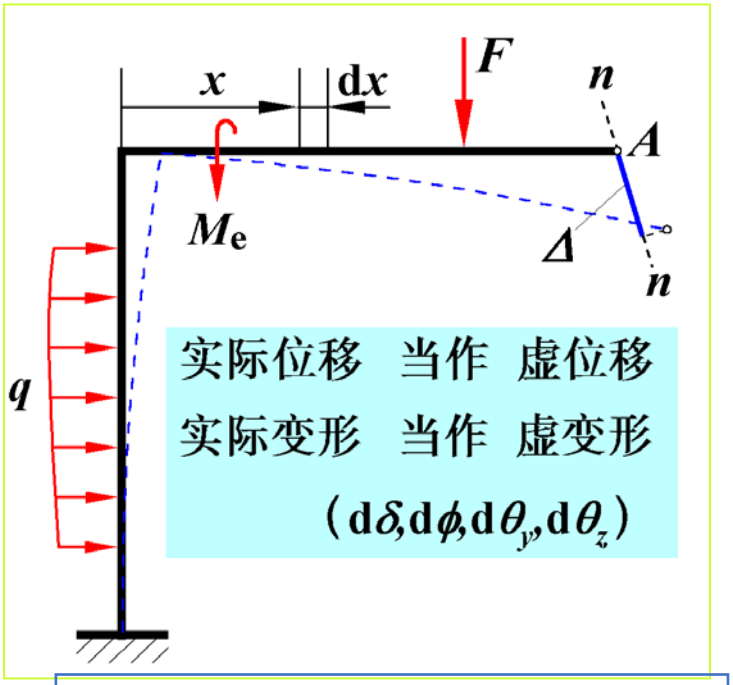

4）思考：

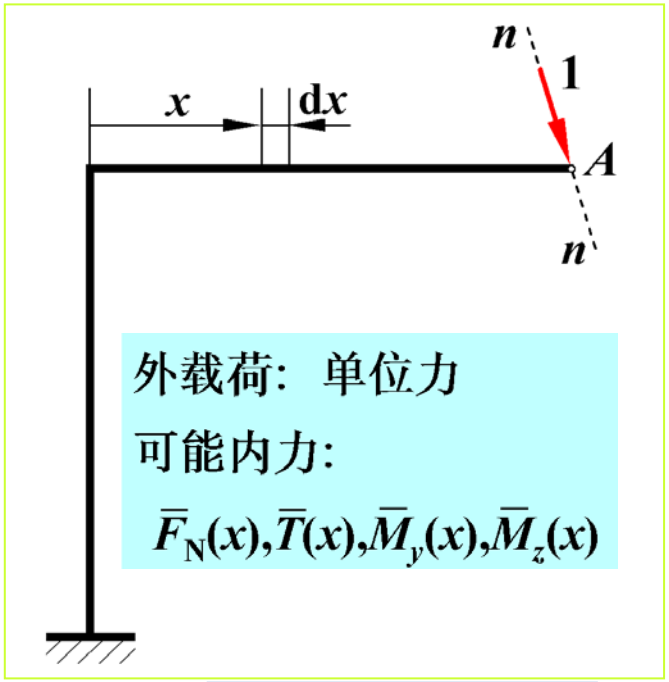



若杆件上作用有集中力或集中力矩，如何证明？

# 3、单位载荷法

结构的载荷状态




构造单位（载荷）状态

$\Delta$ --- $d\delta, d\phi, d\theta_y, d\theta_z$ → 运动许可场

$1$ --- $\bar{F}_N, \bar{T}, \bar{M}_y, \bar{M}_z$ → 静力许可场

$$1\Delta = \int_l \bar{F}_N(x)d\delta + \int_l \bar{T}(x)d\phi + \int_l \bar{M}_y(x)d\theta_y + \int_l \bar{M}_z(x)d\theta_z$$

# 3、单位载荷法

$$\Delta=\int_l \overline{F}_N(x)\mathrm{d}\delta+\int_l \overline{T}(x)\mathrm{d}\phi+\int_l \overline{M}_y(x)\mathrm{d}\theta_y+\int_l \overline{M}_z(x)\mathrm{d}\theta_z$$

*实际变形*由载荷状态下的*实际内力*确定

**关于位移与单位载荷**

**所求位移是单位载荷的相应位移**

$\Delta$ —广义位移，施加相应单位广义载荷

- $\Delta$ - 线位移，加单位力
- $\Delta$ - 角位移，加单位力矩
- $\Delta$ - 相对线位移，加一对相等相反单位力
- $\Delta$ - 相对角位移，加一对相等相反单位力矩

**关于位移方向**

与单位载荷同向为正

**虚力原理（余虚功原理）**

- 单位力为虚力；
- 直接给出单位载荷法。

# 3、单位载荷法

$$\Delta=\int_l\left[\overline{F}_{\mathrm{N}}(x)\mathrm{d}\delta+\overline{T}(x)\mathrm{d}\varphi+\overline{M}_y(x)\mathrm{d}\theta_y+\overline{M}_z(x)\mathrm{d}\theta_z\right]$$

*实际变形* 由载荷状态下的*实际内力* 确定

## 对于线弹性杆或杆系

$$\mathrm{d}\delta=\frac{F_{\mathrm{N}}(x)\mathrm{d}x}{EA},\quad \mathrm{d}\varphi=\frac{T(x)\mathrm{d}x}{GI_{\mathrm{t}}},\quad \mathrm{d}\theta_y=\frac{M_y(x)\mathrm{d}x}{EI_y},\quad \mathrm{d}\theta_z=\frac{M_z(x)\mathrm{d}x}{EI_z}$$

桁架

$$\Delta=\sum_{i=1}^{n}\frac{\overline{F}_{\mathrm{N}i}F_{\mathrm{N}i}l_i}{E_iA_i}$$

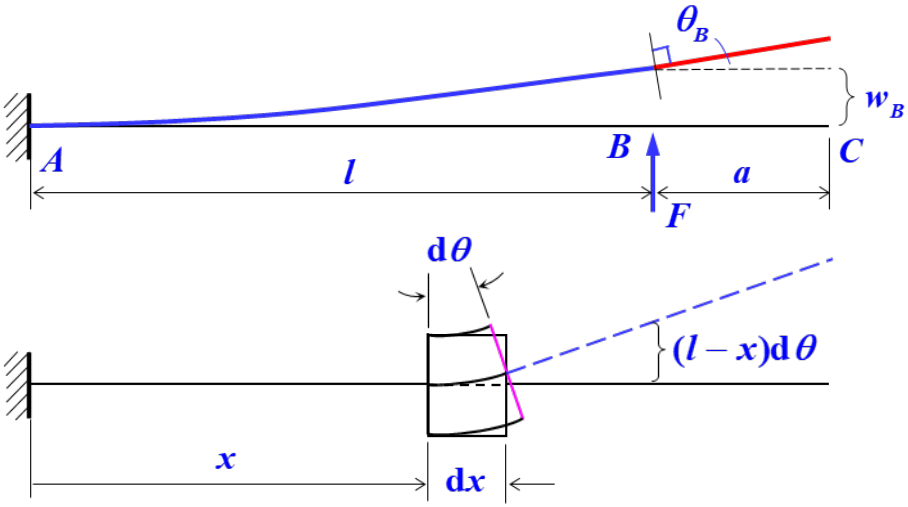
组合变形梁和刚架

$$\Delta=\int_l\frac{\overline{T}(x)T(x)}{GI_{\mathrm{t}}}\mathrm{d}x+\int_l\frac{\overline{M}_y(x)M_y(x)}{EI_y}\mathrm{d}x+\int_l\frac{\overline{M}_z(x)M_z(x)}{EI_z}\mathrm{d}x$$

# 3、单位载荷法

## 影响系数的力学意义

$$\mathrm{d}\theta_B = \mathrm{d}\theta, \quad \mathrm{d}w_B = (l-x)\mathrm{d}\theta$$




**影响系数**——微段变形的贡献

$$\mu_{\theta B} = 1, \quad \mu_{wB} = l - x$$

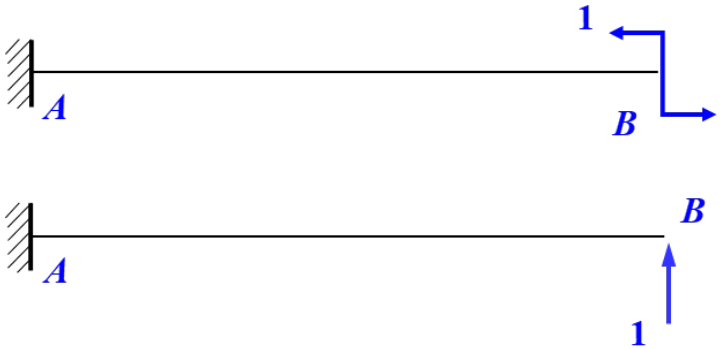
$$\theta_B = \int_0^l \mu_{\theta B}\,\mathrm{d}\theta, \quad w_B = \int_0^l \mu_{wB}\mathrm{d}\theta$$

微段变形累加形成截面挠度与转角

## 构造单位载荷状态

$$\overline{M}_1(x) = 1 \qquad \theta_B = \int_0^l \overline{M}_1(x)\mathrm{d}\theta$$

$$\overline{M}_2(x) = l - x \quad w_B = \int_0^l \overline{M}_2(x)\mathrm{d}\theta$$

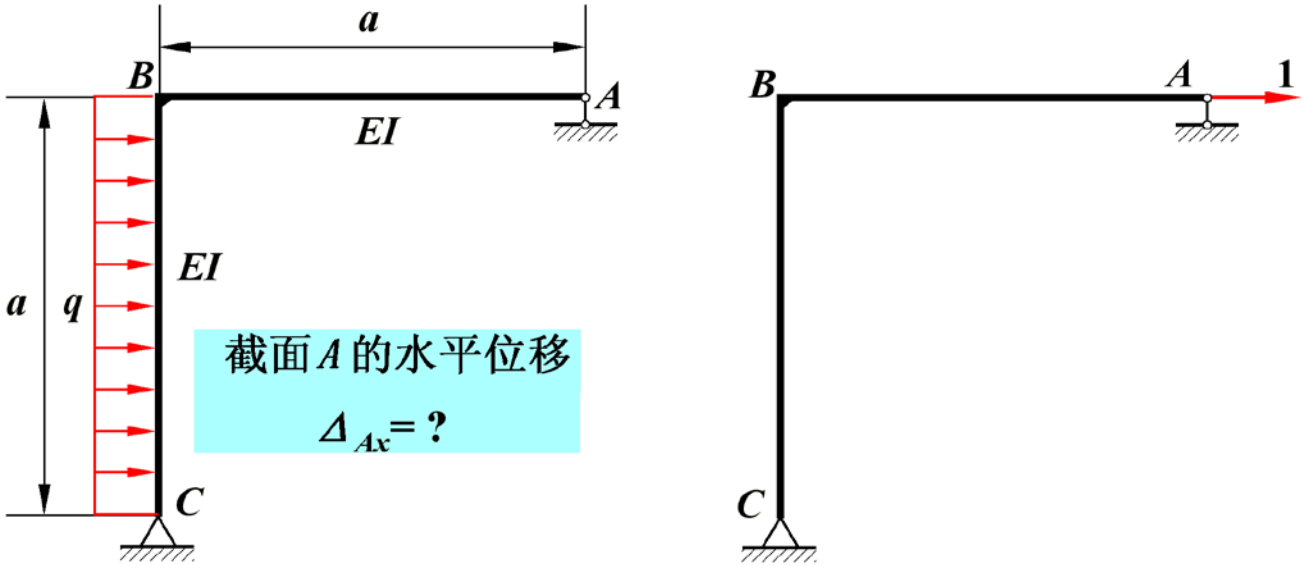



$$\mu_{\theta B} = \overline{M}_1(x), \quad \mu_{wB} = \overline{M}_2(x)$$

单位力引起的内力，
反映了微段变形的影响，与材料无关。

# 4、单位载荷法应用

例13-11 图示刚架，*EI*为常数，求A截面水平位移。



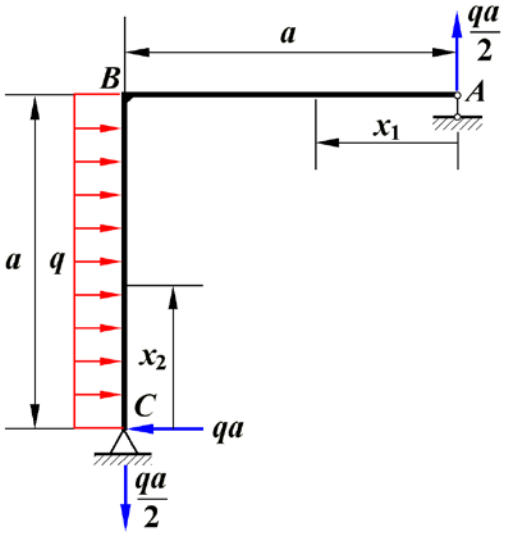

**解：** 1）构造单位载荷状态

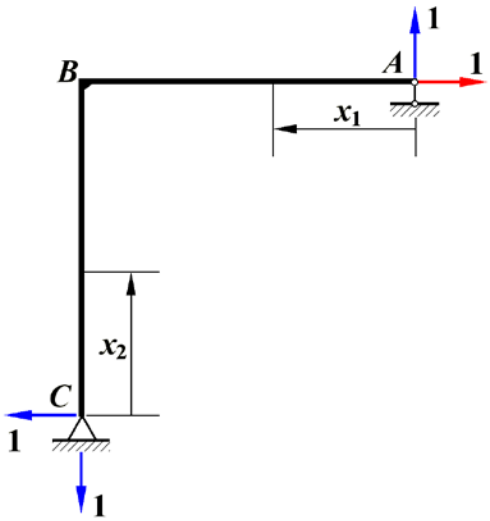
深刻理解相应位移概念，准确施加单位载荷

# 4、单位载荷法应用

2）建立弯矩方程

分段、坐标、符号保持一致




**AB段：** $\overline{M}(x_1)=1\cdot x_1=x_1$， $M(x_1)=\dfrac{qa}{2}x_1$

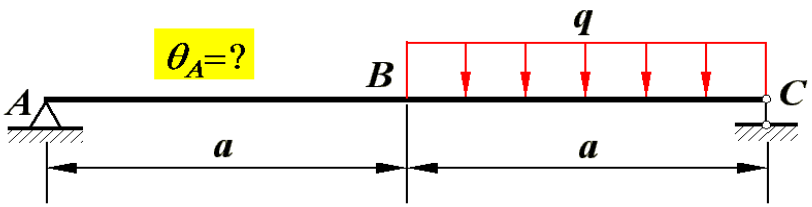
**BC段：** $\overline{M}(x_2)=1\cdot x_2=x_2$， $M(x_2)=qax_2-\dfrac{q}{2}x_2^2$

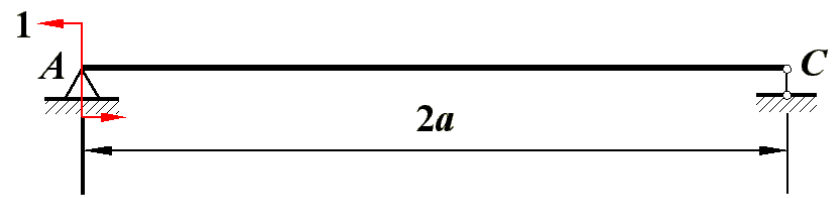
3）求位移

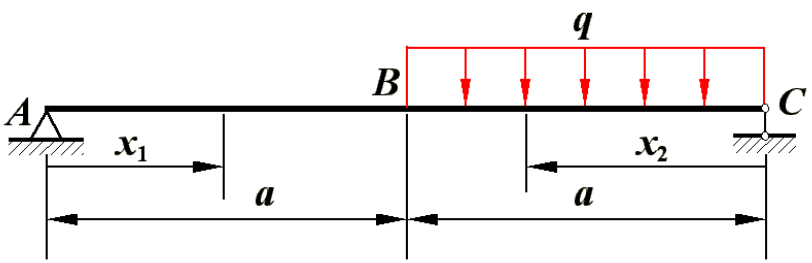
$$\Delta_{Ay}=\frac{1}{EI}\int_0^a x_1\cdot\frac{qa}{2}x_1\mathrm{d}x_1+\frac{1}{EI}\int_0^a x_2\left(qax_2-\frac{q}{2}x_2^2\right)\mathrm{d}x_2=\frac{3qa^4}{8EI}\ (\rightarrow)$$

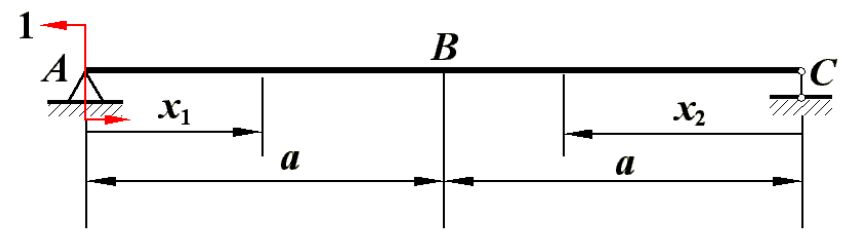
# 4、单位载荷法应用

例13-12 图示简支梁，$EI$为常数，求$\theta_A$.



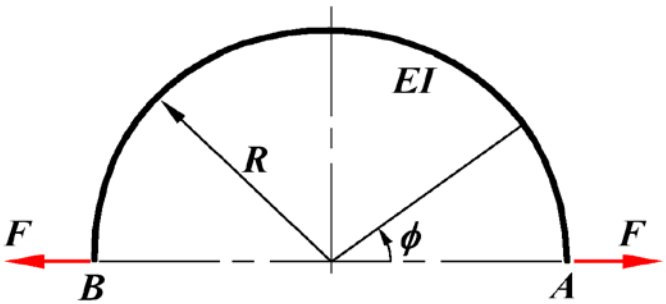




分段、坐标、符号
保持一致

$$\theta_A = \int_l \frac{\overline{M}(x)M(x)}{EI}\,\mathrm{d}x$$

# 4、单位载荷法应用

例13-13 图示1/2圆弧小曲率杆，$EI$为常数，求截面A和B的相对转角$\theta_{A/B}$.



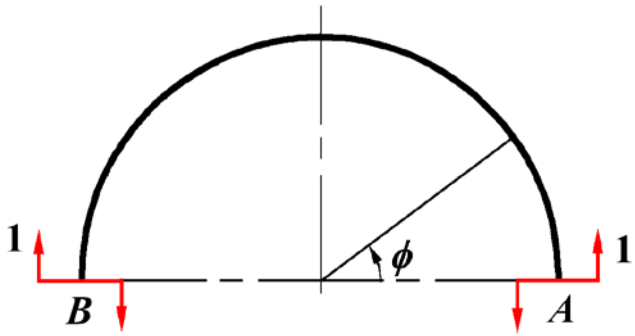

**解：** 施加一对单位力矩

如何理解？

$$\overline{M}(\phi)=-1 \qquad M(\phi)=-FR\sin\phi$$

$$\theta_{A/B}=\frac{1}{EI}\int_0^{\pi}(-1)(-FR\sin\phi)\,R\mathrm{d}\phi$$

$$=\frac{2FR^2}{EI}\quad(\circlearrowright\circlearrowleft)$$




$$\theta_{A/B}=1\times\theta_A+1\times\theta_B=\int_0^{\pi}\overline{M}(\phi)\frac{M(\phi)}{EI}R\mathrm{d}\phi$$

外力虚功

单位载荷法的理论基础是虚功原理

# 4、单位载荷法应用

例13-14 图示平面刚架空间受力，求自由端挠度。

**解：**

*AB*段： $\overline{M}(x_1)=-x_1$

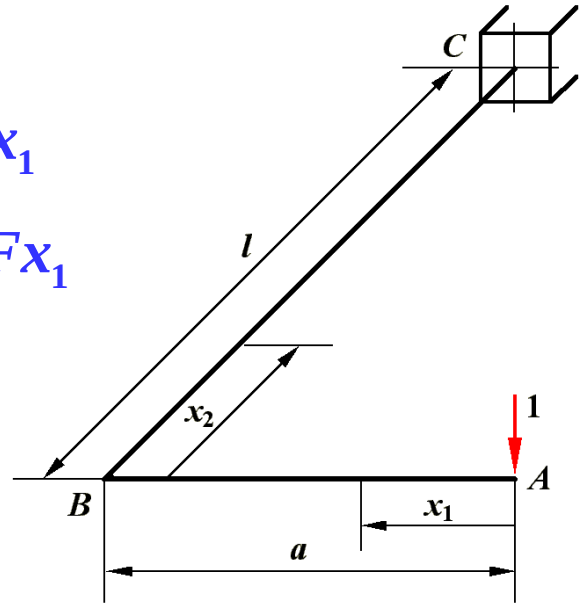
$M(x_1)=-Fx_1$

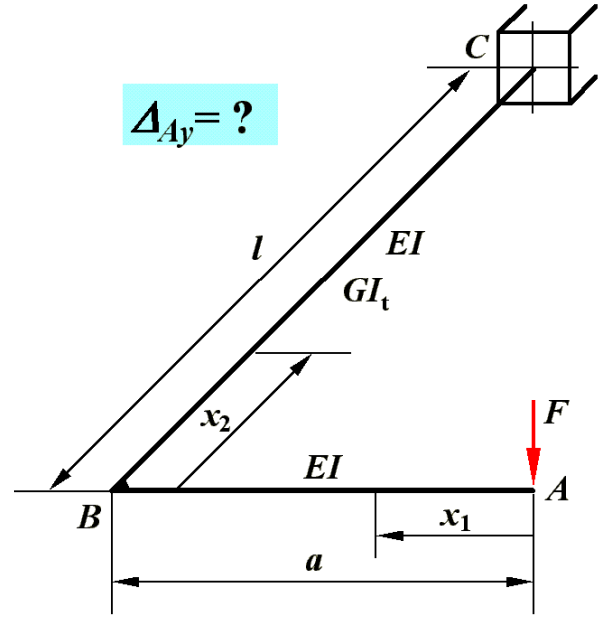
*BC*段：

$\overline{M}(x_2)=-x_2$

$\overline{T}(x_2)=-a$

$M(x_2)=-Fx_2 \quad T(x_2)=-Fa$

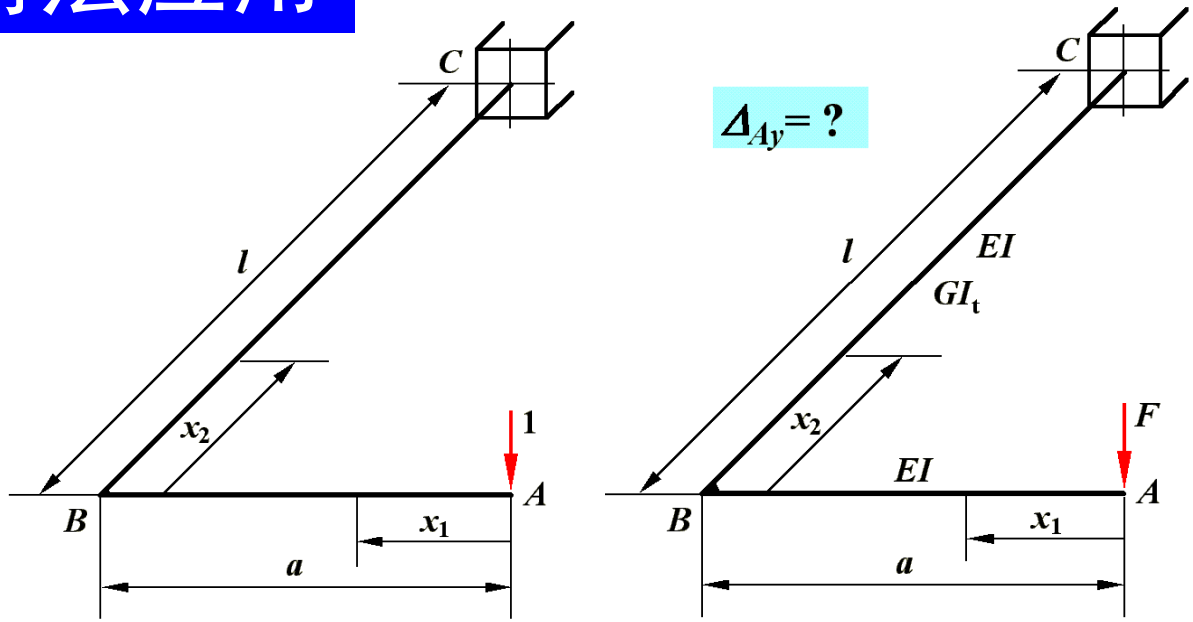



$$\Delta_{Ay}=\int_0^a \frac{\overline{M}(x_1)M(x_1)}{EI}\,\mathrm{d}x_1+\int_0^l \frac{\overline{M}(x_2)M(x_2)}{EI}\,\mathrm{d}x_2+\int_0^l \frac{\overline{T}(x_2)T(x_2)}{GI_t}\,\mathrm{d}x_2$$

# 4、单位载荷法应用

**解：**




$w_B$

$$\Delta_{Ay}=\int_0^a \frac{x_1\cdot Fx_1}{EI}\,\mathrm{d}x_1+\int_0^l \frac{x_2\cdot Fx_2}{EI}\,\mathrm{d}x_2+\int_0^l \frac{a\cdot Fa}{GI_t}\,\mathrm{d}x_2$$

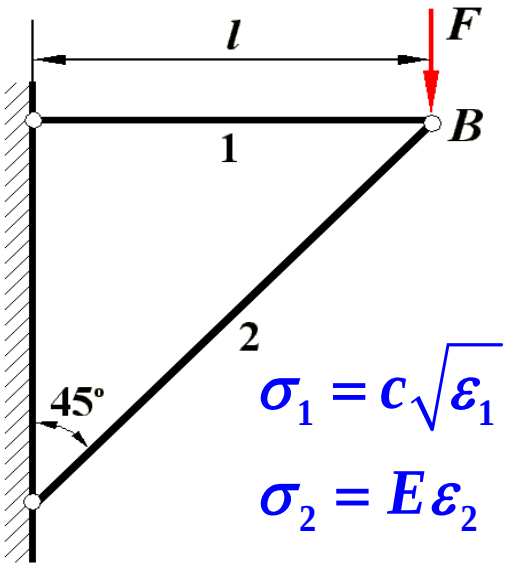
$w_{Ay1}$

$$\Delta_{Ay}=\frac{F(a^3+l^3)}{3EI}+\frac{Fa^2l}{GI_t}\qquad(\downarrow)$$

$\varphi_B a$

# 4、单位载荷法应用

例13-15 图示桁架，二杆截面积均为 $A$，$E$和$c$为材料常数，求$\Delta_{By}$.




$$\sigma_1 = c\sqrt{\varepsilon_1}$$

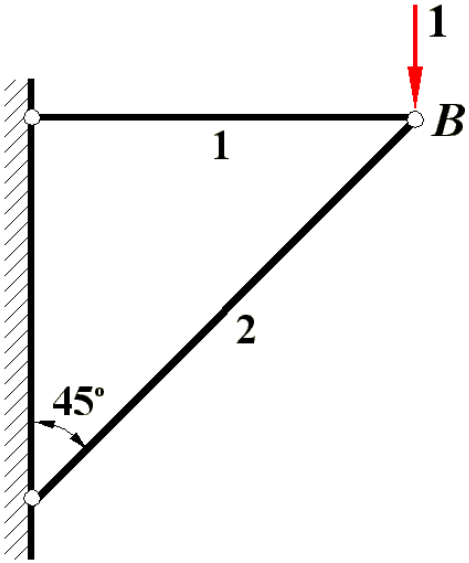
$$\sigma_2 = E\varepsilon_2$$

解：

$$\Delta_{By} = \int_l \overline{F}_{\mathrm{N}}(x)\mathrm{d}\delta = \sum_{i=1}^{2} \overline{F}_{\mathrm{N}i}\Delta l_i$$

$$\overline{F}_{\mathrm{N1}} = 1, \quad \overline{F}_{\mathrm{N2}} = -\sqrt{2};$$

$$F_{\mathrm{N1}} = F, \quad F_{\mathrm{N2}} = -\sqrt{2}F$$




$$\Delta l_2 = \frac{F_{\mathrm{N2}} l_2}{EA} = \frac{(-\sqrt{2}F)\sqrt{2}l}{EA} = -\frac{2Fl}{EA}$$

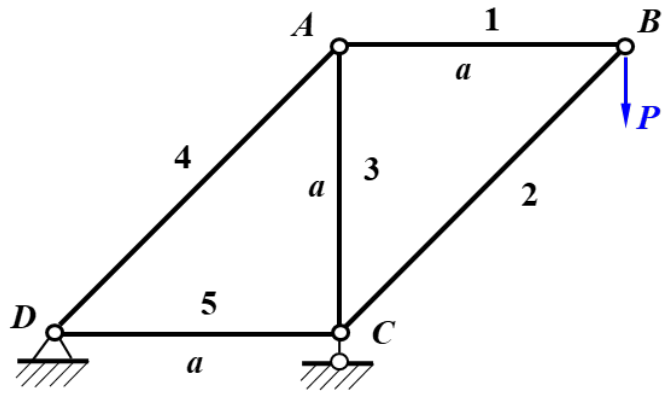
$$\Delta l_1 = \varepsilon_1 l_1 = \frac{\sigma_1^2}{c^2} l_1 = \frac{F_{\mathrm{N1}}^2 l}{A^2c^2} = \frac{F^2 l}{A^2c^2}$$

$$\Delta_{By} = \overline{F}_{\mathrm{N1}}\Delta l_1 + \overline{F}_{\mathrm{N2}}\Delta l_2 = \frac{F^2 l}{A^2c^2} + \frac{2\sqrt{2}Fl}{EA} \quad (\downarrow)$$

# 4、单位载荷法应用

例13-16 图示桁架，各杆$EA$相同，

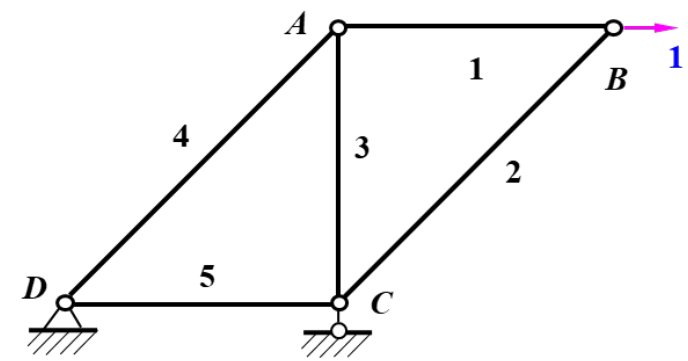
求：$\Delta_B$，$\theta_{AB}$




解：

桁架内力列表

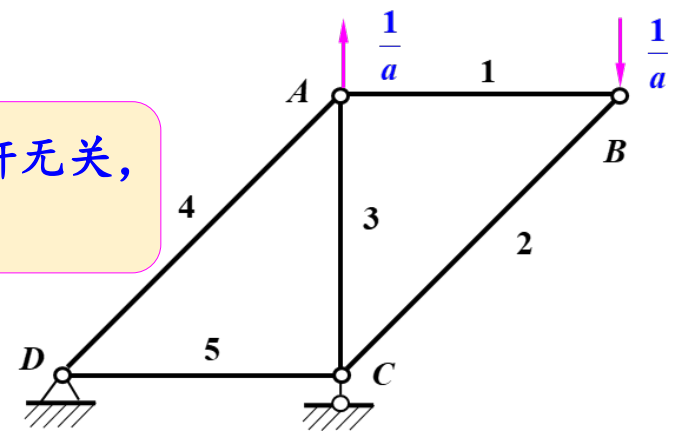
| | $l_i$ | $N_i$ | $\bar{N}_{1i}$ | $\bar{N}_{1i}N_il_i$ | $\bar{N}_{2i}$ | $\bar{N}_{2i}N_il_i$ |
|---|---|---|---|---|---|---|
| 1 | $a$ | $P$ | 1 | $Pa$ | $1/a$ | $P$ |
| 2 | $\sqrt{2}a$ | $-\sqrt{2}P$ | 0 | 0 | $-\sqrt{2}/a$ | $2\sqrt{2}P$ |
| 3 | $a$ | $-P$ | $-1$ | $Pa$ | 0 | 0 |
| 4 | $\sqrt{2}a$ | $\sqrt{2}P$ | $\sqrt{2}$ | $2\sqrt{2}Pa$ | $\sqrt{2}/a$ | $2\sqrt{2}P$ |
| 5 | $a$ | $-P$ | 0 | 0 | $-1/a$ | $P$ |




$$\Delta_B=\sum_{i=1}^{5}\frac{\bar{N}_{1i}N_il_i}{EA}=\frac{2(1+\sqrt{2})Pa}{EA}(\rightarrow)$$

水平位移与2、5杆无关，转动与3杆无关。

$$\theta_{AB}=\sum_{i=1}^{5}\frac{\bar{N}_{2i}N_il_i}{EA}=\frac{2(1+2\sqrt{2})P}{EA}(\circlearrowleft)$$



# 小结：变形体虚功原理

满足平衡方程、静力边界条件

满足变形协调条件、位移边界条件

外力 $q(x)$

内力 $M(x)$

静力许可场

虚位移 $w^*(x)$

虚变形 $\mathrm{d}\theta^*$

运动许可场

彼此独立

外力虚功 $W_e$

内力虚功 $W_i$

适用于线弹性、非线弹性、非弹性体

本讲：虚功原理与单位载荷法

内力虚功等于外力虚功

下讲：冲击应力分析